

Biochemical Experiment Report

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Experimental Title: Extraction and Identification of Genomic DNA from Peripheral Blood Leukocytes

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I. Purpose and Principle

Purpose:

1. To extract genomic DNA from peripheral blood leukocytes of healthy young rabbits.
2. To assess the purity of the extracted DNA using the absorbance ratios at 260 nm/280 nm (A_{260}/A_{280}) and 260 nm/230 nm (A_{260}/A_{230}).

Principle:

Genomic DNA extraction involves the following steps:

1. **Cell Lysis:** Low-salt buffer and detergents (NP-40, SDS) disrupt cell and nuclear membranes to release DNA.
 2. **Protein Removal:** High salt (6 M NaCl) precipitates proteins, which are separated by centrifugation.
 3. **DNA Precipitation:** Ice-cold ethanol dehydrates DNA, causing it to precipitate.
 4. **Washing and Purification:** 70% ethanol removes salts, and TE buffer dissolves the DNA. Purity is determined spectrophotometrically:
 - A_{260}/A_{280} indicates protein contamination.
 - A_{260}/A_{230} reflects organic solvent or salt residues.
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II. Materials and Methods

Reagents:

- 20% NP-40, 10% SDS, 6 M NaCl
- Low-salt buffer (10 mmol/L Tris-HCl, 10 mmol/L KCl, 2 mmol/L EDTA, 4 mmol/L $MgCl_2$)
- Ice-cold ethanol, 70% ethanol, TE buffer

Instruments:

- Micropipettes, centrifuge, NanoDrop 2000 spectrophotometer

Sample Source:

- Whole blood from healthy young adult rabbits (age: X months; sex: X), confirmed disease-free.

Procedure:

1. **Cell Lysis:** 200 μL whole blood mixed with low-salt buffer and NP-40; centrifuged to pellet leukocytes.
 2. **Protein Removal:** SDS lyses nuclei; NaCl precipitates proteins; supernatant (containing DNA) collected.
 3. **DNA Precipitation:** Ice-cold ethanol added to precipitate DNA; pellet washed with 70% ethanol.
 4. **Purity Analysis:** A260, A280, and A230 measured using NanoDrop.
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III. Results

Observations and Raw Data:

1. **Post-Centrifugation:**
 - Step 2: Clear supernatant with white leukocyte pellet.
 - Step 7: Cloudy solution after ethanol addition; white DNA threads formed upon centrifugation.
2. **Spectrophotometry Data:**
 - DNA concentration: **22.75 ng/ μL**
 - $A_{260}/A_{280} = \mathbf{1.29}$
 - $A_{260}/A_{230} = \mathbf{0.35}$

Data Processing:

Direct readings from spectrophotometer; no further calculations required.

IV. Discussion

1. **Result Analysis:**
 - **$A_{260}/A_{280} = 1.29$:** Below the pure DNA range (1.8–2.0), indicating protein contamination (e.g., residual SDS).
 - **$A_{260}/A_{230} = 0.35$:** Far below the ideal range (2.0–2.5), suggesting organic solvent (ethanol) or salt carryover.
2. **Challenges and Anomalies:**

- Slow DNA dissolution: Over-drying or insufficient TE buffer volume.
 - Low A260/A280: Optimize protein removal (e.g., extend centrifugation or increase SDS concentration).
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Detailed Answers to PPT Questions

1. Reagent Functions and Experimental Observations

Question: Explain the roles of the following reagents and describe observations at each step.

- **Low-salt buffer:**
 - **Role:** Maintains osmotic balance during lysis; stabilizes DNA.
 - **Observation:** Homogenized solution; white leukocyte pellet after centrifugation.
- **NP-40:**
 - **Role:** Nonionic detergent for selective membrane lysis.
 - **Observation:** Turbid mixture; pink hemoglobin supernatant post-centrifugation.
- **SDS:**
 - **Role:** Disrupts nuclear membranes; releases DNA.
 - **Observation:** Viscous solution with protein aggregates.
- **6 M NaCl:**
 - **Role:** Salt-induced protein precipitation.
 - **Observation:** Clear supernatant (DNA) and white protein pellet.
- **Ice-cold ethanol:**
 - **Role:** DNA dehydration and precipitation.
 - **Observation:** White DNA threads formed upon inversion.

2. Spectrophotometric DNA Purity Assessment

Question: How to evaluate DNA purity using A260/A280 and A260/A230 ratios?

- **A260/A280:**
 - **Standard:** 1.8–2.0 for pure DNA.
 - **Result (1.29):** Severe protein contamination → increase SDS/proteinase K treatment.
- **A260/A230:**
 - **Standard:** 2.0–2.5 for pure DNA.
 - **Result (0.35):** Organic/salt residues → additional ethanol washes or column purification.
- **Reference:** *"Note on the 260/280 and 260/230 Ratios"* (Superstar APP) highlights salt/solvent interference.

V. Conclusion

Genomic DNA was successfully extracted but with suboptimal purity (A260/A280=1.29; A260/A230=0.35). Key issues:

1. Protein contamination → enhance removal steps (e.g., protease K).
 2. Organic/salt residues → optimize washing (e.g., extended drying).
- The DNA (22.75 ng/μL) can be stored at -20°C but requires further purification for downstream applications.
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